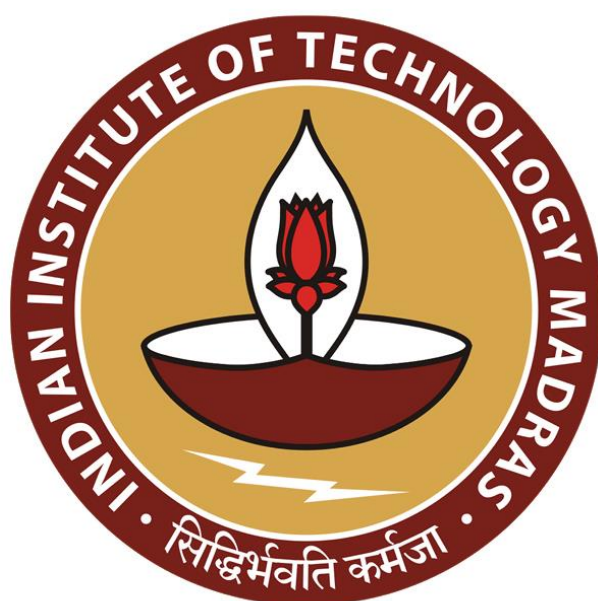




**IIT Madras**  
ONLINE DEGREE

**Programming Concepts Using Java**  
**Online Degree Programme**  
**B. Sc in Programming and Data Science**  
**Diploma Level**  
**Prof. Madhavan Mukund**  
**Department of Computer Science**  
**Chennai Mathematical Institute**

**The Philosophy of OO Programming**

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The screenshot shows a presentation slide with a blue header bar containing the IIT Madras logo and the title 'Algorithms + Data Structures = Programs'. The main content area is white with a bulleted list. A small video inset in the bottom left shows a man in a blue checkered shirt. The bottom of the slide has a blue footer bar with the text 'Madhavan Mukund The philosophy of OO programming Programming Concepts using Java 2/7'.

- Title of Niklaus Wirth's introduction to Pascal
- Traditionally, algorithms come first
- Structured programming
  - Design a set of procedures for specific tasks
  - Combine them to build complex systems
- Data representation comes later
  - Design data structures to suit procedural manipulations

So, before we go into more details about java let us try to understand a little bit why object oriented programming is of interest to us. So, there is a very famous book by Nicholas Worth the inventor of the language Pascal called Algorithms + Data Structures =Programs. So, most people would agree that programming involves both algorithms and data structures.

But the title of this book is interesting because it emphasizes the usual view that algorithms come first. So, when we think about programming we first think about what we want to do in terms of procedures. And in fact in the 1960s and 1970s the focus was on organizing these procedures in something called structured programming. So, the idea is that instead of trying to build a large monolithic program.

You break up your program into smaller units or procedures and then if you have a library of these procedures then when the task at hand requires to be solved you can

build an application using this library of procedures. So, you first have the algorithms encoded as procedures or functions and then since you know that you have these functions and procedures which expect information to come data has to flow in and flow out in a particular format you then choose the data representation for the problem at hand given the procedures. So, you design your data structures to suit the procedural manipulations.

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## Object Oriented design

- Reverse the focus
- First identify the data we want to maintain and manipulate
- Then identify algorithms to operate on the data
- Claim: works better for large systems
- Example: simple web browser
  - 2000 procedures manipulating global data
  - ... vs 100 classes, each with about 20 methods
  - Much easier to grasp the design
  - Debugging: an object is in an incorrect state
  - Search among 20 methods rather than 2000 procedures

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So, the philosophy of object oriented design is to reverse the focus of this. So, here you first identify the data that you want to maintain and manipulate. So, for instance if you are building a banking system you need to talk about customers bank accounts different types of deposits and so on. Then you identify the algorithms that will operate on the data. So, the logic for doing it this way is that there is an argument that it works better for large systems.

So, let us consider a reasonably complicated application something like a web browser. Now for the sake of argument let us assume that a web browser consists of 2000 different functions or procedures and all of these operate on the information that is maintained in the web browser which is some kind of global data about the various web pages that are open and the various things that have happened and the various addresses you have stored and so on.

So, you might reorganize this web browser in an object oriented framework using say hundred classes of objects you might think of a page that is being displayed as an object

and you might have operations to open it to close it to refresh it and so on. So, say these 2000 procedures operating on a large scale global data instead get reorganized as a hundred classes each of them has roughly 20 functions.

So, you still have 2000 procedures or functions but they are organized now in this more manageable framework of 100 classes with 20 methods maybe it might be more than 20 maybe 25 maybe you end up having more procedures but they are localized to the classes. So, one argument is that if you organize it this way because the units that you have to comprehend are smaller it is much easier as a programmer to grasp the design to have a big picture of what is going on across the entire application.

The other part is that debugging becomes a lot easier. So, typically when a program crashes or does something unexpected is because some part of the program is in the incorrect state ok some value is set wrong. So, in object oriented terminology some object in your program is in an incorrect state it has the wrong values now what caused this to happen.

So, if we are in this object oriented framework we know that that particular object is manipulated by the methods associated with it. So, there may be 20 to 30 methods associated with that object and we have to check those to find out what went wrong on the other hand in a classical situation where we had 2000 global procedures we have to examine each of them potentially to see which of them could have done it.

So, in this way narrowing down the scope of an error also becomes easier when you manage your code in this way. So, therefore object oriented design is really a method to try and manage complex code into a more comprehensive framework that a programmer can grasp.

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**Object Oriented design: Example**

- An order processing system typically involves
  - Items
  - Orders
  - Shipping addresses
  - Payments
  - Accounts

Nouns

- What happens to these objects?
  - Items are added to orders
  - Orders are shipped, cancelled
  - Payments are accepted, rejected

Verbs

- Nouns signify objects, verbs denote methods that operate on objects
- Associate with each order, a method to add an item



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So, let us look at a typical example of how you would go about it. So, supposing you are designing a system to process orders say for an online e-commerce website. So, you think about what all happens in this. So, there are some items that people will order there are the orders themselves which consists of people combining different items and then asking to pay for them. So, they will pay for them.

So, there will be payments there will be accounts the there might be some kind of a possibility for the customer to store some data in terms of a prepayment or have a card or something that works on that thing and of course there will be addresses where the object should be shipped. So, you have these different aspects which go into this ordering system and then there are things that you do with these objects right.

So, you have to for instance construct an order by adding items once you have an order ready it may be shipped or it may be cancelled when you have a payment that is made by a customer it may be accepted or for various reasons it may be rejected. So, if you think about this one broad way of classifying it is that you have these nouns right. So, here you have at the top you have nouns right. So, these are things and then you have activities. So, these are verbs. So, the verbs consist of methods that operate on the noun.

So, the nouns are your object and the verbs are your method. So, this is one way of thinking about how to organize an object oriented design right.

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## Designing objects

- Behaviour — what methods do we need to operate on objects?
- State — how does the object react when methods are invoked?
  - State is the information in the instance variables
  - Encapsulation — should not change unless a method operates on it
- Identity — distinguish between different objects of the same class
  - State may be the same — two orders may contain the same item
- These features interact
  - State will typically affect behaviour
  - Cannot add an item to an order that has been shipped
  - Cannot ship an empty order

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So, what happens now when we actually start constructing the system right. So, we have to design these objects. So, what do we have to think about? So, of course the most obvious thing we have to think about is the behavior of each object right. So, what are the kinds of things that we do to an object. So, what are the functions that we need to operate on objects now these functions will manipulate what is called the state of the object.

So, each object remember will have some local variables we call them instance variables. So, the setting of the instance variables is usually called the state of the object. So, it might be in an initial state it might be in a in an order situation for instance you might have an empty order then you might have an order which has been put into the shopping cart and then you might have an order which has been fulfilled and shipped right.

So, the state is the information that you have in the instance variables and one of the fundamental principles of object oriented design and abstract data types in general is that this information should be encapsulated right. So, the information that is inside which is the state of the object should not be available outside except through the methods that you have defined.

So, any change that happens to the state should happen through a method. So, if there is a change of state you can narrow down which methods could have changed the state and try to track down what is going on finally one of the ideas here is that you have these



templates or classes and you have many different objects. So, you need a way to keep track of them right.

So, you need to have ways of identifying these objects from each other and this identification is not necessarily only in terms of state there could be many objects of a given type which are all in the same state two people could order the exact same kind of things. So, if you look at the state of the order it might say that they have exactly the same items which are being ordered but they are still two different orders. So, you have to maintain the identity of objects.

So, that you do not mix them up now these broad features of objects of course are not unconnected. So, the state of an object will typically tell you what you can do with it right. So, the behavior is typically a function of the state. So, when you have an order you typically want to add items to the order but if the order has already been shipped it has been completed and shipped you can no longer add.

So, an order may remain in the system because the customer may want to go back and look up old orders. So, the order may not be cancelled from the system but it has a state in which you cannot change it. So, you cannot add things on the other hand if you have done nothing you have just started and created an empty order then it makes no sense to ship it. So, therefore the operations you can perform on an object will be affected naturally by the state.

So, these are not mutually exclusive things. So, when you talk about behaviour you have to think about state also and identity at the same time because one object may think of talk about the identity of another object.

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- **Dependence**
  - `Order` needs `Account` to check credit status
  - `Item` does not depend on `Account`
  - Robust design minimizes dependencies, or **coupling** between classes
- **Aggregation**
  - `Order` contains `Item` objects
- **Inheritance**
  - One object is a specialized version of another
  - `ExpressOrder` inherits from `Order`
  - Extra methods to compute shipping charges, priority handling



So if you look at the relationship between different classes of objects that we have in our system there could be a dependence right. One type of object may require another type of object or one class may require another class to do its work. For instance if you are placing an order and then this order is going to be paid for by say the wallet of a customer this wallet will be stored in the account. So, the order needs to know something about the account.

So, the class order will depend on the class account on the other hand the individual items which are for sale have information about their price that what type of item they are and. So, on and how much maybe how much you have in stock now clearly an item is going to be in use only if it is part of an order but on its own an item needs to know nothing about a customer's bank account or payment wallet or whatever.

So, items do not depend on accounts. So, we can be sure that if we add if we do something with the accounts implementation nothing will change for item and vice versa and this in general is a principle that we would like to follow not just in this but in every kind of programming but here is particularly easy to think of it this way because we have these different objects and you would like to decouple them right.

So, if you have dependencies between objects then they are supposed to be coupled the coupling between classes. So, you want to minimize the coupling to make your system robust. So, that if you change something in one place it should not change something in

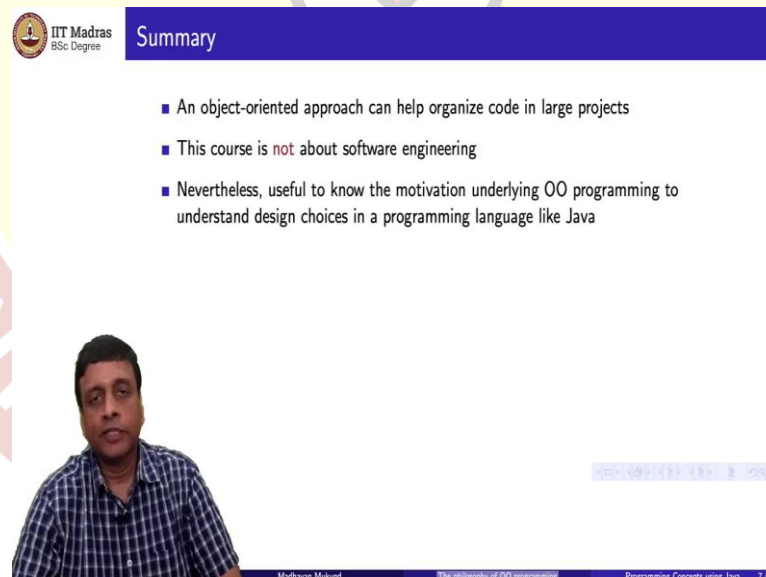


another place of course the other thing that you can do with classes is nest them right. So, what is an order an order consists of items.

So, items will always be added to orders. So, an order object will contain item objects the other thing that we will see as we go along is inheritance we have already talked about this loosely but the main reason that object oriented programming is richer than just abstract data types is the idea that you can have one data type which is an enhanced version of another type right or a specialized version of another type.

So, for instance we have orders but there may be some kind of a way by which a customer can pay extra and ask for quicker delivery. So, we might have something called an express order. So, an express order would be very much like an order it will have all the features of an order it will have items and so on. A shipping address but you will also have extra things about how much extra has to be paid what is the kind of delivery service that is going to be used and so on. So, we have this notion of inheritance where you can take an existing class and enhance it.

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### Summary

- An object-oriented approach can help organize code in large projects
- This course is **not** about software engineering
- Nevertheless, useful to know the motivation underlying OO programming to understand design choices in a programming language like Java

Mathavan Mahand

The philosophy of OO programming

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So, to summarize we want to understand objective the motivation for object-oriented programming because that will drive really our understanding of how a language like java has been designed right. So, the purpose of this course is not to teach you how to build systems using object-oriented program this is not a software engineering course this is a course about programming languages and programming language design.

So, through java we are going to illustrate various features and in particular because java is object oriented but lot of these features will be tied to object oriented programming. So, we are not going to try and emphasize how to write good object oriented programming or what are good naming conventions for variables or how to organize your objects and so on.

But it is good to understand why object object-oriented programming is useful because otherwise some of the design decisions and some of the features available in the language do not really make sense.

